

Natural language statement

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Theorem 1.3.3. Let D be an open subset of $\mathbb{R}^n$ and $(t_0, a_1, a_2, \dots, a_n)$ in D . (a) Assume g in $C(D, \mathbb{R}^n)$. Then there exists $\delta > 0$ such that IVP (1.3.10) has at least one solution which exists for $|t - t_0| \leq \delta$. (b) Assume g in $C(D, \mathbb{R}^n)$, and as a function of $(t, y_1, y_2, \dots, y_n)$, g is locally Lipschitz in $(y_1, y_2, \dots, y_n)$ on D . Then there exists $\delta > 0$ such that IVP (1.3.10) has a unique solution which exists for $|t - t_0| \leq \delta$.

ODE

```
def IsTrajectory {n : N} (F : R -> (Fin n -> R) -> (Fin n -> R))
  (x : R -> (Fin n -> R)) : Prop :=
  forall t : R, HasDerivAt x (F t (x t)) t

def companionField {n : N} (g : R -> (Fin n -> R) -> R) (t : R)
  (y : (Fin n -> R)) : (Fin n -> R) :=
  fun i : Fin n => if h : i.1 + 1 < n then y <i.1 + 1, h> else g t y

def LocallyLipschitzInState {n : N} (D : Set (R x (Fin n -> R)))
  (f : R -> (Fin n -> R) -> (Fin n -> R)) : Prop :=
  forall p in D, exists U in nhds p, exists K : NNReal,
    forall t : R, LipschitzOnWith K (f t) {x | (t, x) in U inter D}

theorem kong_1_3_3_nth_order_scalar_ivp
  {n : N} {D : Set (R x (Fin n -> R))} {g : R -> (Fin n -> R) -> R}
  {t0 : R} {a : (Fin n -> R)} (hD : IsOpen D) (hpoint : (t0, a) in D) :
  (ContinuousOn (fun p : R x (Fin n -> R) => g p.1 p.2) D ->
    exists : R, 0 < /\ let I := Set.Icc (t0 - ) (t0 + )
      exists y : R -> (Fin n -> R), IsTrajectoryOn I (companionField g) y
    /\ y t0 = a /\
      forall t in I, (t, y t) in D) /\
  (ContinuousOn (fun p : R x (Fin n -> R) => g p.1 p.2) D ->
    LocallyLipschitzInState D (companionField g) ->
      exists : R, 0 < /\ let I := Set.Icc (t0 - ) (t0 + )
        exists y : R -> (Fin n -> R), IsTrajectoryOn I (companionField g)
          y /\ y t0 = a /\
            forall t in I, (t, y t) in D) /\
        forall z : R -> (Fin n -> R), IsTrajectoryOn I (companionField
          g) z ->
          z t0 = a -> (forall t in I, (t, z t) in D) -> Set.EqOn z y I)
  := by
  sorry
```

Theorem 1.5.3. Assume that f in $C(D, \mathbb{R}^n)$, df/dx in $C(D, \mathbb{R}^{n \times n})$, and df/dmu in $C(D, \mathbb{R}^{n \times k})$. Then IVP $(V[t_0, x_0, mu])$ has a unique solution $x(t; t_0, x_0, mu)$ which is C^1 in t_0, x_0 , and mu in its domain. Furthermore, let $J(t; t_0, x_0, mu) := df/dx(t, x(t; t_0, x_0, mu); mu)$. Then (a) $(dx/dmu)(t; t_0, x_0, mu)$ is the solution of the IVP $z' = J(t; t_0, x_0, mu)z + df/dmu(t, x(t; t_0, x_0, mu); mu)$, $z(t_0)=0$; (b) $(dx/dx_0)(t; t_0, x_0, mu)$ is the solution of $z' = J(t; t_0, x_0, mu)z$, $z(t_0)=I$; (c) $(dx/dt_0)(t; t_0, x_0, mu)$ is the solution of $z' = J(t; t_0, x_0, mu)z$, $z(t_0)=-f(t_0, x_0; mu)$. Here I stands for the $n \times n$ identity matrix.

ODE

```
def IsTrajectory {n : N} (F : R -> (Fin n -> R) -> (Fin n -> R))
  (x : R -> (Fin n -> R)) : Prop :=
  forall t : R, HasDerivAt x (F t (x t)) t

theorem kong_1_5_3_differentiable_dependence
  {n k : N} {D : Set (R x (Fin n -> R) x (Fin k -> R))}
  {f : R -> (Fin n -> R) -> (Fin k -> R) -> (Fin n -> R)}
  (hD : IsOpen D) (hf : ContDiffOn R 1
    (fun p : R x (Fin n -> R) x (Fin k -> R) => f p.1 p.2.1 p.2.2) D)
  :
  exists (I : R -> (Fin n -> R) -> (Fin k -> R) -> Set R)
  (x : R -> R -> (Fin n -> R) -> (Fin k -> R) -> (Fin n -> R)),
  (forall t0 x0 mu, (t0, x0, mu) in D -> IsOpen (I t0 x0 mu) /\
    (I t0 x0 mu).OrdConnected /\ t0 in I t0 x0 mu /\
    IsTrajectoryOn (I t0 x0 mu) (fun t y => f t y mu) (fun t => x t t0
    x0 mu) /\
    x t0 t0 x0 mu = x0 /\
    (forall t in I t0 x0 mu, (t, x t t0 x0 mu, mu) in D) /\
    forall y, IsTrajectoryOn (I t0 x0 mu) (fun t z => f t z mu) y -> y
    t0 = x0 ->
      (forall t in I t0 x0 mu, (t, y t, mu) in D) ->
        Set.EqOn y (fun t => x t t0 x0 mu) (I t0 x0 mu)) /\
  (let flowDomain := {p : R x R x (Fin n -> R) x (Fin k -> R) |
    p.1 in I p.2.1 p.2.2.1 p.2.2.2};
    ContDiffOn R 1 (fun p => x p.1 p.2.1 p.2.2.1 p.2.2.2) flowDomain) /\
  (forall t0 x0 mu, (t0, x0, mu) in D ->
    let z := fun t => fderiv R (fun eta => x t t0 x0 eta) mu
    z t0 = 0 /\ forall t in I t0 x0 mu, HasDerivAt z
      ((fderiv R (fun y => f t y mu) (x t t0 x0 mu)).comp (z t) +
        fderiv R (fun eta => f t (x t t0 x0 mu) eta) mu) t) /\
  (forall t0 x0 mu, (t0, x0, mu) in D ->
    let z := fun t => fderiv R (fun y => x t t0 y mu) x0
    z t0 = ContinuousLinearMap.id R (Fin n -> R) /\ forall t in I t0 x0
    mu,
      HasDerivAt z ((fderiv R (fun y => f t y mu) (x t t0 x0 mu)).comp
        (z t)) t) /\
  forall t0 x0 mu, (t0, x0, mu) in D ->
    let z := fun t => deriv (fun s => x t s x0 mu) t0
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		<pre> z t0 = -f t0 x0 mu /\ forall t in I t0 x0 mu, HasDerivAt z ((fderiv R (fun y => f t y mu) (x t t0 x0 mu)) (z t)) t := by sorry </pre>
Theorem 2.3.1 (Variation of Parameters Formula). Let $X(t)$ be a fundamental matrix solution of Eq. (H) and t_0 in (a,b). Then the general solution of Eq. (NH) is $x = X(t)c + \int_{t_0}^t X(t)X^{-1}(s)f(s) ds$. In particular, the solution of the IVP consisting of Eq. (NH) and the IC $x(t_0)=x_0$ is $x = X(t)X^{-1}(t_0)x_0 + \int_{t_0}^t X(t)X^{-1}(s)f(s) ds$.	ODE	<pre> def IsTrajectory {n : N} (F : R -> (Fin n -> R) -> (Fin n -> R)) (x : R -> (Fin n -> R)) : Prop := forall t : R, HasDerivAt x (F t (x t)) t def FundamentalMatrixSolution {n : N} (I : Set R) (A : R -> Matrix (Fin n) (Fin n) R) (X : R -> Matrix (Fin n) (Fin n) R) : Prop := (forall t in I, HasDerivAt X (A t * X t) t) /\ forall t in I, IsUnit (X t) theorem kong_2_3_1_variation_of_parameters {n : N} {I : Set R} {A : R -> Matrix (Fin n) (Fin n) R} {f : R -> (Fin n -> R)} {X : R -> Matrix (Fin n) (Fin n) R} {t0 : R} (hI : I.OrdConnected) (hA : ContinuousOn A I) (hf : ContinuousOn f I) (hX : FundamentalMatrixSolution I A X) (ht0 : t0 in I) : (forall y : R -> (Fin n -> R)), IsTrajectoryOn I (fun t x => A t *v x + f t) y <-> exists c : Fin n -> R, forall t in I, y t = X t *v c + integral s in t0..t, (X t * (X s)^-1) *v f s) /\ forall x0, exists y : R -> (Fin n -> R), y t0 = x0 /\ IsTrajectoryOn I (fun t x => A t *v x + f t) y /\ (forall t in I, y t = (X t * (X t0)^-1) *v x0 + integral s in t0..t, (X t * (X s)^-1) *v f s) /\ forall z : R -> (Fin n -> R), z t0 = x0 -> IsTrajectoryOn I (fun t x => A t *v x + f t) z -> Set.EqOn z y I := by sorry </pre>
Theorem 2.5.3 (Floquet Theorem). Let $X(t)$ be a fundamental matrix solution of Eq. (H-p). Then there exists an R in $C^n \times n$ and a nonsingular omega-periodic P in $C(I, C^n \times n)$ such that $X(t) = P(t)e^{iRt}$.	ODE	<pre> def FundamentalMatrixSolution {n : N} (I : Set R) (A : R -> Matrix (Fin n) (Fin n) R) (X : R -> Matrix (Fin n) (Fin n) R) : Prop := (forall t in I, HasDerivAt X (A t * X t) t) /\ forall t in I, IsUnit (X t) def PeriodicLinearEquation {n : N} (omega : R) (A : R -> Matrix (Fin n) (Fin n) R) : Prop := forall t, A (t + omega) = A t theorem kong_2_5_3_floquet_theorem {n : N} {omega : R} {A : R -> Matrix (Fin n) (Fin n) R} {X : R -> Matrix (Fin n) (Fin n) R} (homega : 0 < omega) (hper : PeriodicLinearEquation omega A) (hA : Continuous A) (hX : FundamentalMatrixSolution univ A X) : exists R : Matrix (Fin n) (Fin n) C, exists P : R -> Matrix (Fin n) (Fin n) C, (forall i j, ContDiff R 1 fun t => P t i j) /\ (forall t, P (t + omega) = P t) /\ (forall t, IsUnit (P t)) /\ forall t, (X t).map (algebraMap R C) = P t * NormedSpace.exp (t * R) := by sorry </pre>
Theorem 3.2.3. Let $\mu_i, i = 1, \dots, n$, be the characteristic multipliers of Eq. (H-p). Then (a) Equation (H-p) is uniformly stable $\Rightarrow \mu_i \leq 1, i = 1, \dots, n$, and $\mu_i = 1$ occurs only when the μ_i's are in the diagonal Jordan block of the transition matrix V; (b) Equation (H-p) is asymptotically stable $\Rightarrow \mu_i < 1, i = 1, \dots, n$; (c) Equation (H-p) is unstable $\Rightarrow$ there exists an i in $\{1, \dots, n\}$ such that either $\mu_i > 1$, or $\mu_i = 1$ which occurs when μ_i is not in the diagonal Jordan block of the transition matrix V.	ODE	<pre> def IsTrajectory {n : N} (F : R -> (Fin n -> R) -> (Fin n -> R)) (x : R -> (Fin n -> R)) : Prop := forall t : R, HasDerivAt x (F t (x t)) t def CharacteristicMultipliers {n : N} (V : Matrix (Fin n) (Fin n) C) (mu : Fin n -> C) : Prop := Matrix.charpoly V = prod i, (Polynomial.X - Polynomial.C (mu i)) def InDiagonalJordanBlock {n : N} (V : Matrix (Fin n) (Fin n) C) (mu : C) : Prop := forall v : (Fin n -> C), (V - mu * (1 : Matrix (Fin n) (Fin n) C)) *v ((V - mu * (1 : Matrix (Fin n) (Fin n) C)) *v v) = 0 -> (V - mu * (1 : Matrix (Fin n) (Fin n) C)) *v v = 0 </pre>

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		<pre> def UniformlyStableLinearEquation {n : N} (A : R -> Matrix (Fin n) (Fin n) R) : Prop := UniformlyStableZeroSolution fun t x => A t *v x def AsymptoticallyStableLinearEquation {n : N} (A : R -> Matrix (Fin n) (Fin n) R) : Prop := AsymptoticallyStableZeroSolution fun t x => A t *v x def UnstableLinearEquation {n : N} (A : R -> Matrix (Fin n) (Fin n) R) : Prop := UnstableZeroSolution fun t x => A t *v x theorem kong_3_2_3_characteristic_multiplier_stability {n : N} {A : R -> Matrix (Fin n) (Fin n) R} {mu : Fin n -> C} {V : Matrix (Fin n) (Fin n) C} {omega : R} (homega : 0 < omega) (hperiodic : PeriodicLinearEquation omega A) (hV : IsPeriodTransitionMatrix omega A V) (hmu : CharacteristicMultipliers V mu) : (UniformlyStableLinearEquation A <-> forall i, mu i <= 1 /\ (mu i = 1 -> InDiagonalJordanBlock V (mu i))) /\ (AsymptoticallyStableLinearEquation A <-> forall i, mu i < 1) /\ (UnstableLinearEquation A <-> exists i, 1 < mu i /\ (mu i = 1 /\ not InDiagonalJordanBlock V (mu i))) := by sorry </pre>
Theorem 3.4.2. Assume that there exists a function p in $C([0, \infty), [0, \infty))$ such that $\int_0^\infty p(t) dt < \infty$, and $r(t, x) \leq p(t) x$ for sufficiently small x and all t in $[0, \infty)$. (a) If Eq. (H) is uniformly stable, then the zero solution of Eq. (3.4.6) is uniformly stable. (b) If Eq. (H) is uniformly stable and asymptotically stable, then the zero solution of Eq. (3.4.6) is uniformly stable and asymptotically stable.	ODE	<pre> def IsTrajectory {n : N} (F : R -> (Fin n -> R) -> (Fin n -> R)) (x : R -> (Fin n -> R)) : Prop := forall t : R, HasDerivAt x (F t (x t)) t def IntegrableSmallPerturbation {n : N} (p : R -> R) (r : R -> (Fin n -> R) -> (Fin n -> R)) : Prop := ContinuousOn p (Set.Ici 0) /\ (forall t, 0 <= p t) /\ IntegrableOn p (Set.Ici 0) /\ exists rho : R, 0 < rho /\ forall t, 0 <= t -> forall x, x < rho -> r t x <= p t * x def UniformlyStableLinearEquation {n : N} (A : R -> Matrix (Fin n) (Fin n) R) : Prop := UniformlyStableZeroSolution fun t x => A t *v x def AsymptoticallyStableLinearEquation {n : N} (A : R -> Matrix (Fin n) (Fin n) R) : Prop := AsymptoticallyStableZeroSolution fun t x => A t *v x def UniformlyStableZeroSolution {n : N} (F : R -> (Fin n -> R) -> (Fin n -> R)) : Prop := forall eps : R, 0 < eps -> exists delta : R, 0 < delta /\ forall t0 x, 0 <= t0 -> IsTrajectory F x -> x t0 < delta -> forall t, t0 <= t -> x t < eps def AsymptoticallyStableZeroSolution {n : N} (F : R -> (Fin n -> R) -> (Fin n -> R)) : Prop := UniformlyStableZeroSolution F /\ exists delta : R, 0 < delta /\ forall t0 x, 0 <= t0 -> IsTrajectory F x -> x t0 < delta -> Tendsto x atTop (nhds 0) theorem kong_3_4_2_integrable_perturbation_stability {n : N} {A : R -> Matrix (Fin n) (Fin n) R} {r : R -> (Fin n -> R) -> (Fin n -> R)} {p : R -> R} (hr : IntegrableSmallPerturbation p r) : (UniformlyStableLinearEquation A -> UniformlyStableZeroSolution (fun t x => A t *v x + r t x)) /\ (UniformlyStableLinearEquation A -> AsymptoticallyStableLinearEquation A -> AsymptoticallyStableZeroSolution (fun t x => A t *v x + r t x)) := by sorry </pre>
Theorem 3.5.2. Let $D = \{x \text{ in } \mathbb{R}^n : x \leq l\}$ for some $l > 0$ and	ODE	<pre> def IsTrajectory {n : N} (F : R -> (Fin n -> R) -> (Fin n -> R)) </pre>

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V in $C^1(D, \mathbb{R})$. Assume that $V(x)$ is positive definite and $V(x)$ is negative semi-definite. Moreover, if the set $D_0 := \{x \in D : V(x) = 0\}$ does not contain any nontrivial orbit of Eq. (A), then the zero solution of Eq. (A) is uniformly stable and asymptotically stable.

```
(x : R -> (Fin n -> R)) : Prop :=
forall t : R, HasDerivAt x (F t (x t)) t

def LyapunovFunctionOnBall {n : N} (l : R) (V : (Fin n -> R) -> R)
(F : (Fin n -> R) -> (Fin n -> R)) : Prop :=
ContDiffOn R 1 V (Metric.closedBall 0 l) /\ V 0 = 0 /\
(forall x in Metric.closedBall (0 : (Fin n -> R)) l, x != 0 -> 0 < V x)
/\
forall x in Metric.closedBall (0 : (Fin n -> R)) l,
fderivWithin R V (Metric.closedBall 0 l) x (F x) <= 0

def NontrivialOrbitInZeroDerivativeSet {n : N} (l : R)
(V : (Fin n -> R) -> R) (F : (Fin n -> R) -> (Fin n -> R)) : Prop :=
forall x : R -> (Fin n -> R), IsAutonomousTrajectory F x ->
(forall t, x t in Metric.closedBall (0 : (Fin n -> R)) l /\
fderivWithin R V (Metric.closedBall 0 l) (x t) (F (x t)) = 0) ->
x = 0

def UniformlyStableZeroSolution {n : N}
(F : R -> (Fin n -> R) -> (Fin n -> R)) : Prop :=
forall eps : R, 0 < eps -> exists delta : R, 0 < delta /\ forall t0 x,
0 <= t0 -> IsTrajectory F x -> ||x t0|| < delta -> forall t, t0 <= t ->
||x t|| < eps

def AsymptoticallyStableZeroSolution {n : N}
(F : R -> (Fin n -> R) -> (Fin n -> R)) : Prop :=
UniformlyStableZeroSolution F /\ exists delta : R, 0 < delta /\ forall t0
x,
0 <= t0 -> IsTrajectory F x -> ||x t0|| < delta -> Tendsto x atTop
(nhds 0)

theorem kong_3_5_2_lasalle_invariance_stability
{n : N} {l : R} {F : (Fin n -> R) -> (Fin n -> R)} {V : (Fin n -> R) ->
R}
(hl : 0 < l) (hF : ContDiff R 1 F) (hF0 : F 0 = 0)
(hV : LyapunovFunctionOnBall l V F)
(horbit : NontrivialOrbitInZeroDerivativeSet l V F) :
AsymptoticallyStableZeroSolution (fun _ x => F x) := by
sorry
```

Theorem 4.5.3. Let $x(t)$ be a solution of system (A-2) and Γ its orbit. Assume Γ^{++} is contained in a compact set E subset $\mathbb{R}^2$ and system (A-2) has at most a finite number of equilibria in E . Then one of the following four statements is true: (a) $\Omega(\Gamma^{++})$ contains only one equilibrium of system (A-2); (b) Γ is a closed orbit; (c) $\Omega(\Gamma^{++})$ is a closed orbit; (d) $\Omega(\Gamma^{++})$ is a graphic for System (A-2). The same conclusion holds when Γ^{++} and $\Omega(\Gamma^{++})$ are replaced by Γ^- and $A(\Gamma^-)$, respectively.

ODE

```
def IsAutonomousTrajectory {n : N} (F : (Fin n -> R) -> (Fin n -> R))
(x : R -> (Fin n -> R)) : Prop :=
IsTrajectory (fun _ => F) x

def omegaLimitSet {n : N} (x : R -> (Fin n -> R)) : Set (Fin n -> R) :=
{y | exists t : N -> R, Tendsto t atTop atTop /\ Tendsto (fun j => x (t
j)) atTop (nhds y)}

def alphaLimitSet {n : N} (x : R -> (Fin n -> R)) : Set (Fin n -> R) :=
{y | exists t : N -> R, Tendsto t atTop atBot /\ Tendsto (fun j => x (t
j)) atTop (nhds y)}

def IsClosedOrbit {n : N} (F : (Fin n -> R) -> (Fin n -> R))
(x : R -> (Fin n -> R)) : Prop :=
IsAutonomousTrajectory F x /\ (exists t, x t != x 0) /\
exists T : R, 0 < T /\ forall t, x (t + T) = x t

def GraphicForPlanarSystem (F : (Fin 2 -> R) -> (Fin 2 -> R))
(Gamma : Set (Fin 2 -> R)) : Prop :=
IsConnected Gamma /\
exists (m : N) (e : Fin m -> (Fin 2 -> R)) (x : Fin m -> R -> (Fin 2 ->
R)),
(forall i, F (e i) = 0) /\ (forall i, IsAutonomousTrajectory F (x i)) /\
(forall i, exists j k, Tendsto (x i) atBot (nhds (e j)) /\ Tendsto (x
i) atTop (nhds (e k))) /\
Gamma = range e union union i, range (x i)

theorem kong_4_5_3_generalized_poincare_bendixson
{F : (Fin 2 -> R) -> (Fin 2 -> R)} {x : R -> (Fin 2 -> R)}
{E : Set (Fin 2 -> R)}
(hF : ContDiff R 1 F) (hcompact : IsCompact E) (horbit :
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		<pre> IsAutonomousTrajectory F x) (hfinite : {x in E F x = 0}.Finite) : let classify := fun limitSet : Set (Fin 2 -> R) => (exists e, limitSet = {e} /\ F e = 0) \/ IsClosedOrbit F x \/ (exists y, IsClosedOrbit F y /\ limitSet = range y) \/ GraphicForPlanarSystem F limitSet ((forall t, 0 <= t -> x t in E) -> classify (omegaLimitSet x)) /\ ((forall t, t <= 0 -> x t in E) -> classify (alphaLimitSet x)) := by sorry </pre>
Theorem 5.4.2. Assume (5.4.2) and (5.4.4) hold with $\alpha'(0) = 0$. Then either (a) all orbits of system (5.4.1-0) in a neighborhood of (0,0) are closed orbits and system (5.4.1-μ) does not have closed orbits for $\mu \neq 0$ in a neighborhood of $\mu = 0$, or (b) for $\mu > 0$ sufficiently close to zero only or for $\mu < 0$ sufficiently close to zero only, system (5.4.1-μ) has a unique limit cycle $\Gamma(\mu)$ satisfying $\Gamma(\mu) \rightarrow (0,0)$ with its period $T(\mu) \rightarrow 2\pi/\beta$ as $\mu \rightarrow 0$.	ODE	<pre> def IsAutonomousTrajectory {n : N} (F : (Fin n -> R) -> (Fin n -> R)) (x : R -> (Fin n -> R)) : Prop := IsTrajectory (fun _ => F) x def IsClosedOrbit {n : N} (F : (Fin n -> R) -> (Fin n -> R)) (x : R -> (Fin n -> R)) : Prop := IsAutonomousTrajectory F x /\ (exists t, x t != x 0) /\ exists T : R, 0 < T /\ forall t, x (t + T) = x t noncomputable def linearizationMatrix (F : (Fin 2 -> R) -> R -> (Fin 2 -> R)) (mu : R) : (Matrix (Fin 2) (Fin 2) R) := fun i j => fderiv R (fun x => F x mu) 0 (Pi.single j 1) i theorem kong_5_4_2_hopf_friedrich_dichotomy {F : (Fin 2 -> R) -> R -> (Fin 2 -> R)} {beta : R} (hbeta : 0 < beta) (hF : ContDiff R inf (fun p : (Fin 2 -> R) x R => F p.1 p.2) /\ (forall mu, F 0 mu = 0) /\ Matrix.trace (linearizationMatrix F 0) = 0 /\ Matrix.det (linearizationMatrix F 0) = beta ^ 2 /\ HasDerivAt (fun mu => Matrix.trace (linearizationMatrix F mu)) 0 0) : let center := (exists eps : R, 0 < eps /\ forall x : R -> (Fin 2 -> R), IsAutonomousTrajectory (fun y => F y 0) x -> x 0 < eps -> (exists t, x t != x 0) -> IsClosedOrbit (fun y => F y 0) x) /\ exists eps : R, 0 < eps /\ forall mu, 0 < mu -> mu < eps -> not exists x : R -> (Fin 2 -> R), IsClosedOrbit (fun y => F y mu) x /\ x 0 < eps let hopf := fun positiveSide => exists eps : R, exists orbit : R -> R -> (Fin 2 -> R), exists period : R -> R, 0 < eps /\ (forall mu, 0 < mu -> mu < eps -> (if positiveSide then 0 < mu else mu < 0) -> IsClosedOrbit (fun y => F y mu) (orbit mu) /\ 0 < period mu /\ (forall t, orbit mu (t + period mu) = orbit mu t) /\ forall y, IsClosedOrbit (fun z => F z mu) y -> y 0 < eps -> range y = range (orbit mu)) /\ Tendsto (fun mu => orbit mu 0) (nhds[!]= 0) (nhds 0) /\ Tendsto period (nhds[!]= 0) (nhds (2 * Real.pi / beta)) center /\ hopf true /\ hopf false := by sorry </pre>
Theorem 6.6.4. SLP (S-L), (P) has a countably infinite number of eigenvalues $\lambda_{n,i}$, n in $\mathbb{N}_0$, which are all real, bounded below and unbounded above, and can be arranged to satisfy the coupling relations with μ_n and ν_n for n in $\mathbb{N}_0$: $\mu_0 \leq \lambda_{0,0} < \mu_0, \nu_1 < \lambda_{0,1} \leq \{\mu_1, \nu_2\} \leq \lambda_{0,2} < \{\mu_2, \nu_3\} < \lambda_{0,3} \leq \{\mu_3, \nu_4\} \leq \lambda_{0,4} < \dots < \{\mu_n, \nu_{2n}\} < \lambda_{0,2n+1} \leq \{\mu_{2n+1}, \nu_{2n+2}\} \leq \lambda_{0,2n+2} < \dots$. Furthermore, (a) $\lambda_{0,0}$ is geometrically simple; and for $n \geq 1$, $\lambda_{n,i}$ may be geometrically simple or double, and $\lambda_{n,i}$ is geometrically double $\Rightarrow \lambda_{n,i} = \mu_{i,j}$ for some i,j in $\mathbb{N}_0$. (b) The eigenfunctions associated with $\lambda_{0,0}$ have no zeros in $[a,b]$, and for n in $\mathbb{N}_0$, the eigenfunctions associated with $\lambda_{n,i}$ and $\lambda_{n,i+1}$ have exactly $2n+2$ zeros in the half-open interval $[a,b]$.	ODE	<pre> def IsSturmLiouvilleEigenfunction (p q w : R -> R) (a b eigVal : R) (boundary : (R -> R) -> (R -> R) -> Prop) (y : R -> R) : Prop := (exists x in Set.Icc a b, y x != 0) /\ exists y' : R -> R, boundary y y' /\ (forall x in Set.Icc a b, HasDerivWithinAt y (y' x) (Set.Icc a b) x) /\ forall x in Set.Icc a b, HasDerivWithinAt (fun t => p t * y' t) ((q x - eigVal * w x) * y x) (Set.Icc a b) x def PeriodicSturmLiouvilleData (p q w : R -> R) (a b : R) : Prop := a < b /\ ContinuousOn p (Set.Icc a b) /\ ContinuousOn q (Set.Icc a b) /\ ContinuousOn w (Set.Icc a b) /\ (forall x in Set.Icc a b, 0 < p x /\ 0 < w x) def periodicBoundary (p : R -> R) (a b : R) (y y' : R -> R) : Prop := </pre>

Natural language statement

Domain

GroundTruth

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y a = y b /\ p a * y' a = p b * y' b

def dirichletBoundary (a b : R) (y _y' : R -> R) : Prop := y a = 0 /\ y b =
0

def neumannBoundary (a b : R) (_y y' : R -> R) : Prop := y' a = 0 /\ y' b =
0

theorem kong_6_6_4_periodic_sturm_liouville_coupling
{p q w : R -> R} {a b : R}
(hSL : PeriodicSturmLiouvilleData p q w a b) :
exists lam mu nu : N -> R, (forall n, lam n <= lam (n + 1)) /\ Tendsto
lam atTop atTop /\
nu 0 <= lam 0 /\
(forall n, lam (2 * n) < mu (2 * n) /\ lam (2 * n) < nu (2 * n + 1) /\
mu (2 * n) < lam (2 * n + 1) /\ nu (2 * n + 1) < lam (2 * n + 1) /\
lam (2 * n + 1) <= mu (2 * n + 1) /\ lam (2 * n + 1) <= nu (2 * n +
2) /\
mu (2 * n + 1) <= lam (2 * n + 2) /\ nu (2 * n + 2) <= lam (2 * n +
2)) /\
(forall eigVal, (exists y, IsSturmLiouvilleEigenfunction p q w a b
eigVal
(periodicBoundary p a b) y) <=> exists n, lam n = eigVal) /\
(forall eigVal, (exists y, IsSturmLiouvilleEigenfunction p q w a b
eigVal
(dirichletBoundary a b) y) <=> exists n, mu n = eigVal) /\
(forall eigVal, (exists y, IsSturmLiouvilleEigenfunction p q w a b
eigVal
(neumannBoundary a b) y) <=> exists n, nu n = eigVal) /\
(forall y, IsSturmLiouvilleEigenfunction p q w a b (lam 0)
(periodicBoundary p a b) y -> {x in Set.Icc a b | y x = 0} = empty)
/\
(forall y1 y2, IsSturmLiouvilleEigenfunction p q w a b (lam 0)
(periodicBoundary p a b) y1 ->
IsSturmLiouvilleEigenfunction p q w a b (lam 0)
(periodicBoundary p a b) y2 ->
exists c : R, Set.EqOn y2 (c * y1) (Set.Icc a b)) /\
(forall n, (exists i j, lam n = mu i /\ lam n = nu j) <=>
exists y1 y2, IsSturmLiouvilleEigenfunction p q w a b (lam n)
(periodicBoundary p a b) y1 /\
IsSturmLiouvilleEigenfunction p q w a b (lam n)
(periodicBoundary p a b) y2 /\
not exists c : R, Set.EqOn y2 (c * y1) (Set.Icc a b)) /\
forall n, (forall y, IsSturmLiouvilleEigenfunction p q w a b (lam (2
* n + 1))
(periodicBoundary p a b) y ->
{x in Set.Ico a b | y x = 0}.ncard = 2 * n + 2) /\
forall y, IsSturmLiouvilleEigenfunction p q w a b (lam (2 * n + 2))
(periodicBoundary p a b) y ->
{x in Set.Ico a b | y x = 0}.ncard = 2 * n + 2 := by
sorry

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